Financial Fraud Detection

Project Report

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Transaction analysis, fraud detection, customer behaviour and account risk

A four-page interactive Power BI dashboard project

1 Project Overview

The Financial Fraud Detection project combines four Power BI dashboards into one structured reporting solution. The dashboards examine transaction volumes, fraudulent and legitimate activity, customer behaviour, payment channels, devices, prior fraud indicators and account-balance anomalies. The purpose is to help a reviewer locate concentrations of risk and identify transactions that deserve further investigation.

The pages use distinct datasets. Their transaction counts and fraud rates must therefore be interpreted within each page rather than summed or compared as if they represented the same population.

Objectives

- **Monitor fraudulent and legitimate transactions and assess the share of confirmed fraud.**
- **Compare fraud activity by location, transaction type, payment method and device.**
- **Investigate customer behaviour, previous fraud history and spending patterns.**
- **Visualise account balance changes and possible transaction anomalies.**
- **Provide interactive filters and clear business insights for risk reviews.**

2 Problem Statement

Digital payments create large volumes of heterogeneous transaction data. Manual reviews can overlook unusual behaviour, while treating every large transaction as suspicious can lead to misleading results. Business users need a consistent view of transaction counts, monetary exposure and descriptive risk indicators, with clear separation between confirmed fraud, legitimate transactions and cases awaiting review.

3 Data Description

The supplied PDF export documents four dashboards. Figures below are read from visible dashboard cards and are rounded where the display uses K, M or bn. Exact underlying records, source-file coverage and definitions should be confirmed in the Power BI model.

Dashboard	Displayed population	Selected indicators
Page 01: Transaction overview	100K transactions	1K fraud; 1.00% fraud rate; total amount 249.71M
Page 02: Customer behaviour	5K transactions	482 fraud; 9.64% rate; ~4K customers
Page 03: Device intelligence	50K transactions	~16K fraud; 32.13% rate; ~9K unique users
Page 04: Banking anomalies	10.13K transactions	68 confirmed fraud; 2 pending; 0.67% rate

Monetary values are shown without a currency unit because the dashboard export does not establish a consistent currency.

4 Technology Stack

Layer	Tools	Purpose / Evidence
Source data	Excel datasets	Transaction, customer, device and banking records used across the four dashboard pages.
Preparation	Power Query	Data type checks, missing-value handling, status cleansing and transaction-row identifiers.
Reporting	Microsoft Power BI Desktop	Interactive visuals, KPI cards, slicers and dashboards visible in the export.
Measures	DAX	Counts, fraud rates, transaction sums and averages used in report visuals.
Documentation	Word and PDF	Editable narrative and presentation-ready report export.

No machine-learning training pipeline, classifier metrics, code repository or live deployment is verified by the supplied dashboard export. This is documented as a business intelligence and descriptive analytics project.

5 System Architecture

The logical project flow shown below summarises the reporting process; exact relationships and refresh settings are not established by the static export.

Source	Transformation	Analytics output
Excel / transaction datasets	Power Query: data types, cleaning, categories	Power BI model + DAX measures
Fraud flags, customers, accounts, dates	Transaction-level identifiers and calculations	Four reports, KPI cards, charts and slicers

Figure 1 Conceptual data-to-dashboard architecture

Model Design Considerations

- **Keep source tables for different pages logically distinct unless valid shared keys are available.**
- **Use unique transaction identifiers for transaction-level scatter plots.**
- **Distinguish confirmed fraud from unreviewed records before calculating fraud rate.**
- **Use explicit dates and consistent aggregation grains for trend reporting.**

6 Methodology

6.1 Data preparation

Transaction datasets were imported into Power BI. The reported workflow included checking data types and amount fields, reviewing blanks and category values, cleaning text fields, and retaining account identifiers for transaction analysis. For Page 04, the cleaned fraud-status field distinguishes Safe, Fraud, Not reviewed, and any missing or unknown statuses.

6.2 Data modelling and DAX

Business measures calculate total transactions, confirmed fraud cases, safe and unreviewed transactions, confirmed fraud rate, transaction amount and average transaction amount. The confirmed fraud rate on Page 04 uses reviewed cases (Fraud + Safe) as the denominator; the donut chart may instead display shares of total records. These bases should be labelled to avoid confusion.

6.3 Exploratory analysis

The dashboards compare fraud across transaction types, time periods, locations, device categories, card types, payment methods and account-age groups. Scatter charts support the review of amounts against account balance or spending. These are descriptive views and do not independently establish causation or predict future fraud.

6.4 Visual development

Power BI column and bar charts, donut charts, line charts, treemaps, scatter plots and KPI cards were combined. Date, transaction type, fraud status, device and location slicers support page-specific exploration.

6.5 Data validation and quality checks

- **Verify KPI counts and chart totals under identical filters.**
- **Check the meaning and denominator of each fraud-rate measure.**
- **Inspect missing transaction-type, account or fraud-status values instead of deleting them automatically.**
- **Check labels and time granularity: day-based charts must not be presented as monthly trends.**
- **Confirm that account-balance scatter points are shown at the intended transaction grain.**
- **Ensure that synthetic or sample data is not described as verified real-world fraud incidence.**

The PDF export can substantiate visible measures and visuals, but it does not expose the full query logic, DAX formulas, relationships or slicer behaviour.

7 Dashboard Walkthrough and Insights

7.1 Fraud Monitoring & Transaction Overview

The first dashboard summarises a large transaction population and compares fraudulent activity across locations and transaction types. A transaction trend and a location treemap complement the summary metrics.

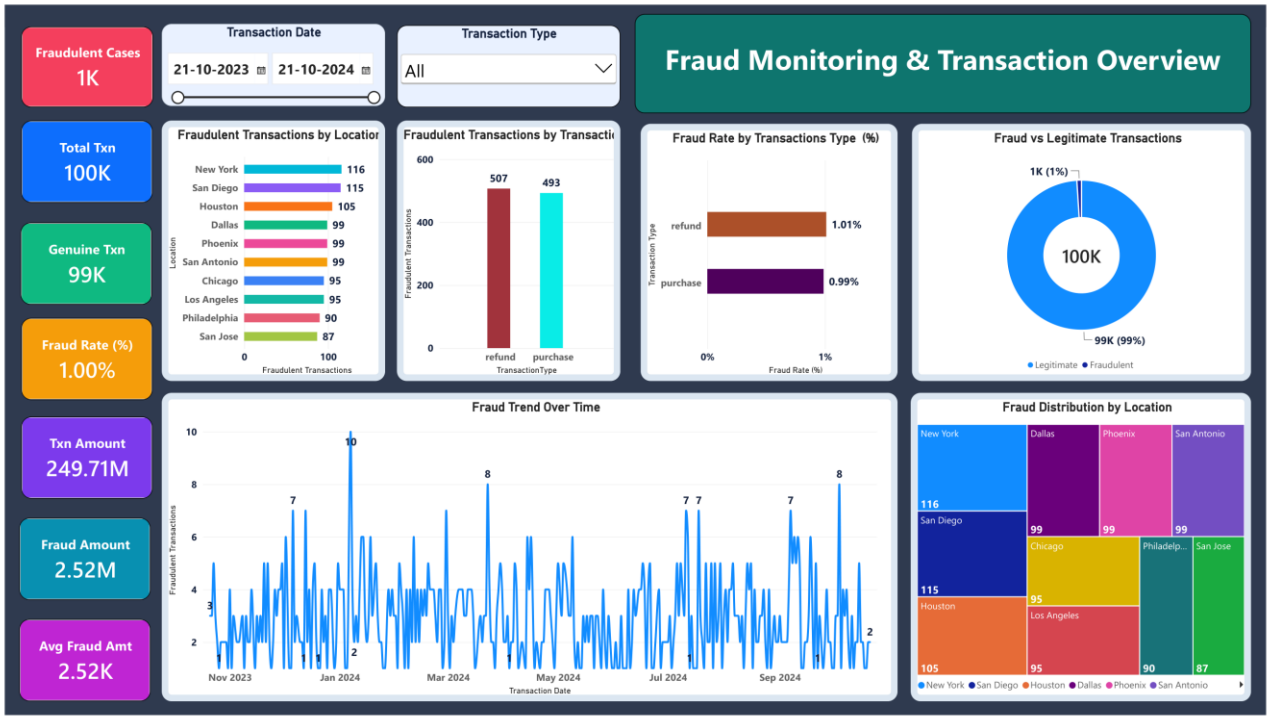


Figure 2 Fraud Monitoring & Transaction Overview | Source: Fraud Detection.pdf, dashboard page 1

Main observations

- **The page displays 100K total transactions, 1K fraudulent cases and 99K legitimate cases, corresponding to a 1.00% fraud rate.**
- **Total transaction amount is displayed as 249.71M, with fraud amount of 2.52M and average fraud amount of 2.52K.**
- **New York (116) and San Diego (115) lead the listed location fraud counts; refund and purchase counts are 507 and 493.**

Interpretation and review

Location counts identify where observed fraud cases are concentrated, not necessarily where fraud risk is highest: rates also depend on total transactions by location. Verify the selected date range before interpreting trend peaks.

7.2 Customer Behaviour & Fraud Risk Analysis

The second dashboard explores customer transactions, regional classification, device usage, monthly variation, account age, merchant categories and payment methods.



Figure 3 Customer Behaviour & Fraud Risk Analysis | Source: Fraud Detection.pdf, dashboard page 2

Main observations

- **Cards display 5K transactions, approximately 4K unique customers, 482 fraudulent transactions and a 9.64% fraud rate.**
- **Fraud amount is displayed as 39.52K; average transaction amount is 79.18.**
- **The international/domestic chart shows 319 domestic versus 163 international fraudulent cases.**
- **Debit Card (103), UPI (100), PayPal (98), Credit Card (96) and NetBanking (85) appear in the payment-method fraud counts.**

Interpretation and review

The device donut and some merchant-category charts present amount totals, not necessarily fraud-case counts; their titles and measure definitions should be reconciled. Older account groups show more recorded cases but require transaction-volume denominators before risk comparisons.

7.3 Advanced Fraud Risk and Device Intelligence

The third dashboard studies device and card activity, historical fraud indicators, merchant categories, daily transaction frequency and transaction amount against account balance.

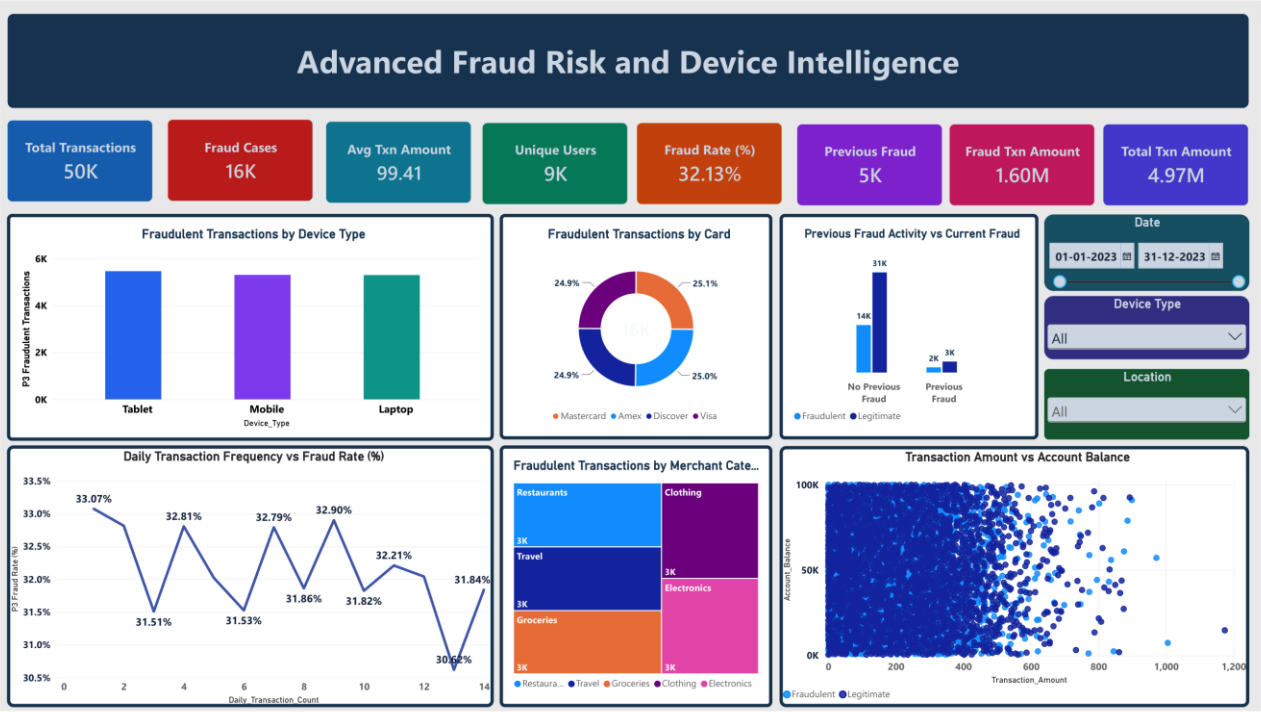


Figure 4 Advanced Fraud Risk and Device Intelligence | Source: Fraud Detection.pdf, dashboard page 3

Main observations

- **The page displays 50K total transactions, approximately 16K fraudulent cases and a 32.13% fraud rate.**
- **Other headline metrics include ~9K unique users, 99.41 average amount, 1.60M fraud transaction amount and 4.97M total amount.**
- **Tablet, Mobile and Laptop have visually similar fraudulent case totals. Card categories are also distributed close to one quarter each.**
- **Prior-fraud-history segmentation and a dense account-balance scatter support deeper exploratory risk review.**

Interpretation and review

A fraud share near 32% is unusually high for many real-world settings and may reflect an intentionally synthetic or enriched dataset; the underlying sampling method is not established from the PDF. The frequency-versus-rate line is not a chronological daily time series.

7.4 Banking Fraud & Account Anomaly Analysis

The fourth dashboard combines reviewed banking statuses with transaction types, transaction timing, geographic labels and account-balance changes.

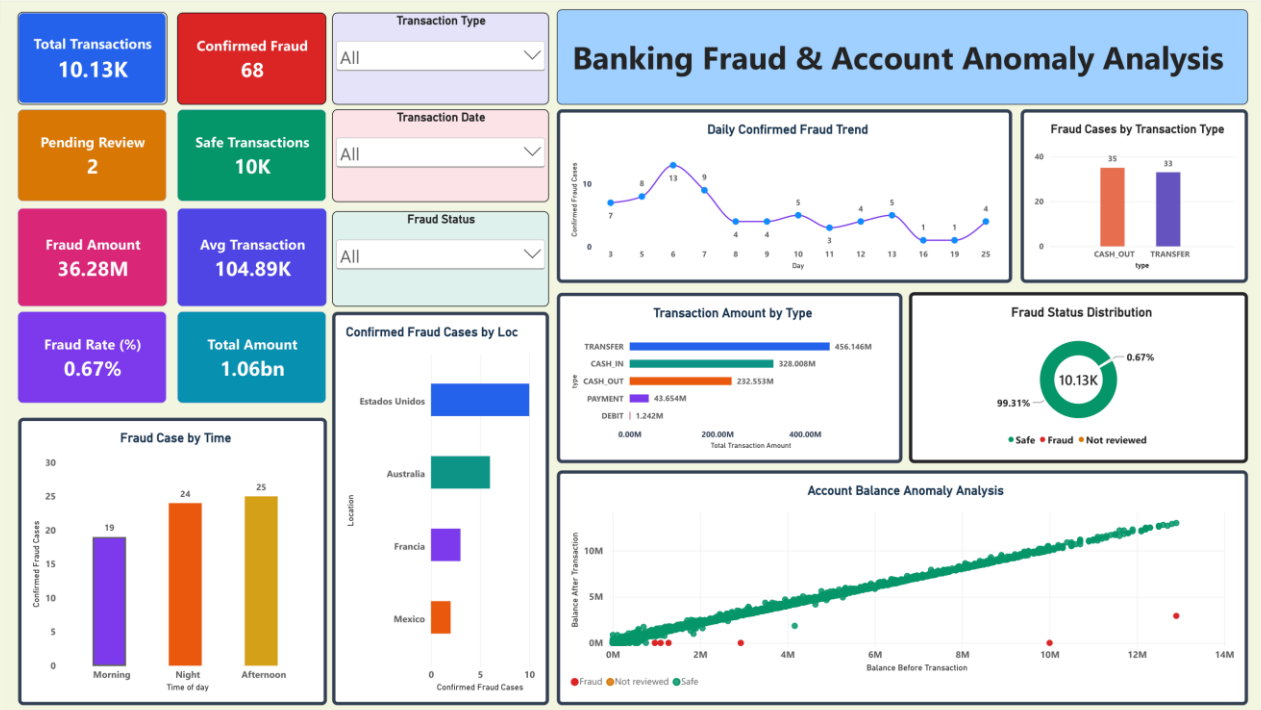


Figure 5 Banking Fraud & Account Anomaly Analysis | Source: Fraud Detection.pdf, dashboard page 4

Main observations

- **The cards show 10.13K total transactions, 68 confirmed fraud cases, two pending reviews and a displayed 0.67% confirmed fraud rate.**
- **Fraud transaction amount is 36.28M, total amount approximately 1.06bn and average transaction amount 104.89K.**
- **Fraud counts by time of day show Morning 19, Night 24 and Afternoon 25, which sum to 68.**
- **The transaction-type fraud chart shows CASH_OUT 35 and TRANSFER 33, also summing to 68.**
- **The balance scatter marks fraudulent cases separately from safe transactions and reveals several visually atypical points.**

Interpretation and review

The location categories are country names in the displayed chart, so "location" is more precise than "branch" unless the data dictionary defines otherwise. Confirm the meaning of balance-before and balance-after changes before identifying individual points as anomalous.

8 Key Insights and Business Recommendations

The four dashboard pages support descriptive fraud monitoring across different analytical populations. They should not be used to calculate one combined fraud rate because the datasets differ in size and likely origin.

Priority	Recommendation	Expected benefit
High	Maintain clear classifications for confirmed fraud, safe transactions and pending review.	Reliable fraud-rate interpretation and review prioritisation.
High	Verify the grain and definitions of transaction-level scatter charts.	Avoid misleading aggregated anomalies.
High	Investigate transaction types and times with notable observed fraud volumes.	Focus manual review resources on useful descriptive signals.
Medium	Compare location, device and payment fraud counts with their total exposure.	Fairer risk comparisons using rates as well as raw counts.
Medium	Create data-quality checks for missing accounts, categories and statuses.	Stronger refresh quality and trust in dashboards.
Medium	Use drill-through views and tooltips for case review.	Faster exploration without cluttered visuals.

Cross-page observations

- **The first dashboard shows a 1.00% fraud rate in 100K transactions.**
- **The second page shows 9.64% across its separate 5K record population.**
- **The third shows 32.13% across a separate 50K dataset.**
- **The fourth shows 68 confirmed cases out of approximately 10.13K records, with two still unreviewed.**

9 Challenges Faced

The following challenges are evident in the supplied dashboards or relate directly to their documentation:

- **Different transaction populations: KPIs and fraud rates across pages cannot be added or directly compared as one sample.**
- **Status definitions: pending or unreviewed transactions must remain separate from confirmed Safe or Fraud categories.**
- **Display rounding: KPI labels in K/M/bn can obscure exact counts and small monetary categories.**
- **Time granularity: one trend uses calendar day values, so it must be described as daily rather than monthly.**
- **Visual density: the third dashboard scatter contains heavy overplotting, reducing interpretability.**
- **Category interpretation: location labels in the fourth dashboard are countries; naming must follow the data dictionary.**
- **Evidence gaps: the dashboard PDF cannot verify source-level data lineage, model relationships, refresh results or ML performance.**

Quality assurance checklist

Check	Acceptance criterion
KPI count integrity	Component fraud counts reconcile to total under the same filters.
Fraud-rate meaning	Reviewed-case or all-record denominator stated unambiguously.
Trend labels	X-axis date grain matches the chart title.
Currency units	Amounts and display units consistent and explained.
Scatter uniqueness	Each intended case uses its unique transaction identifier.
Accessible formatting	Readable labels and sufficient contrast in slides and exports.

10 Conclusion and Future Scope

The Financial Fraud Detection project demonstrates a practical, four-page Power BI reporting framework for transaction monitoring and fraud-risk exploration. It combines high-level KPI monitoring with category breakdowns, customer behaviour, device indicators and account-balance investigation. Interactive slicers make it easier to focus on particular time periods and risk dimensions.

The strongest result is a consistent analytical workflow that makes recorded transaction and fraud patterns accessible to business users. This work is primarily descriptive analytics; it does not demonstrate the accuracy of a deployed fraud prediction model. Validated measures, clear dataset boundaries and carefully defined status categories are essential before using the outputs for operational decisions.

Future scope

- **Document all source datasets, relationships and measure definitions in a data dictionary.**
- **Introduce rule-based exception flags and case drill-through for abnormal balance changes.**
- **Add fraud rates by location, payment method, device and transaction type alongside counts.**
- **Build a controlled model evaluation process if a predictive classifier is later developed.**
- **Add access controls, scheduled refresh and monitored data-quality alerts where deployment permits.**
- **Improve visual accessibility and maintain consistent currency/date formatting across dashboards.**

11 Project Links

Verified repository, deployment and demonstration links were not supplied with the four-page PDF. Add them below if required for submission.

Deliverable	Link / status
GitHub repository	Not provided
Live Power BI report	Not provided
Demo video	Not provided
Project source files	Not attached as part of this report

Sources and editable files

Dashboard evidence: Fraud Detection.pdf (4 pages), provided by the project author. Formatting and section-structure reference: Project_FORESIGHT_Report(2).pdf (18 pages). KPI figures and screenshots derive from the four Fraud Detection dashboards. This editable Word file contains text and tables; dashboard screenshots are embedded images and should be replaced after any Power BI updates.